

NACS 645 – Is visual processing modular

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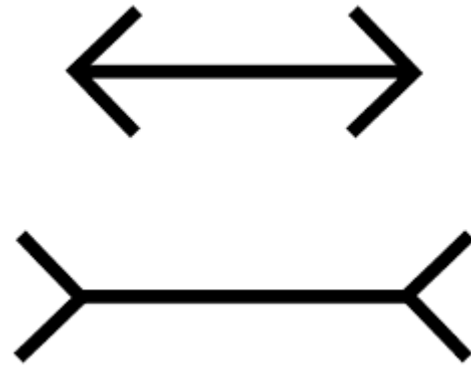
DEPARTMENT OF
PSYCHOLOGY



PROGRAM IN
NEUROSCIENCE &
COGNITIVE SCIENCE

Impenetrability of the perception

Müller-Lyer illusion



Despite knowing the illusion, its effect persists on human perception. Why?

Because low-level (early visual) information operates independently from higher-level information.

Perception is impenetrable to Cognition. Low-level information is encapsulated within early visual modules.

Modularity paradigm

Fodor (1983) distinguished modular and non-modular systems. Modular systems fulfill certain properties:

1. Domain specificity
2. Mandatory operation
3. Limited central accessibility
4. Fast processing
5. Informational encapsulation
6. 'Shallow' outputs
7. Fixed neural architecture
8. Characteristic and specific breakdown patterns
9. Characteristic ontogenetic pace and sequencing

Modular systems are encapsulated : unaffected by other cognitive domains (cognitively impenetrable) and no information is added (non inferential).

Modularist views

Some later researchers/philosophers started drawing a border between perception and cognition.

Pylyshyn (1999) didn't but developed from the modularist view.

- Information encapsulation is an important feature of (early) vision. Ties to cognitive impenetration
- Top-down/complex processing explained by “*local vision-specific memory embodied in early vision*”
- Hence, Pylyshyn's vision of early visual processing is tied to modularity but not fully modularist. Additionally, there is no border drawn between perception and cognition

Fodor's (1983) modularity

1. **Domain specificity:** A modular system processes a **restricted class of inputs and problems**.
2. **Mandatory operation:** Once presented with the appropriate stimulus, **the system automatically engages and runs** to completion **without conscious control**.
3. **Limited central accessibility:** The intermediate **representations** generated within the system are **opaque to consciousness and cannot be accessed for explicit report**.
4. **Fast processing:** Modular systems **operate rapidly**, often on the order of hundreds of milliseconds, reflecting efficiency gained from restricted informational access.
5. **Informational encapsulation:** A module's **operations draw only on its proprietary database and input signals**, remaining insulated from information elsewhere in the mind.
6. **'Shallow' outputs:** The system produces relatively **simple, general, or basic-level representations that are computationally cheap**, rather than highly abstract or theoretical concepts.
7. **Fixed neural architecture:** Modular systems are **implemented in dedicated and circumscribed neural circuits** that can be selectively impaired without disrupting other functions.
8. **Characteristic and specific breakdown patterns:** **Damage** to a module **produces distinctive deficits** (e.g., prosopagnosia, dyslexia) **without widespread cognitive collapse**.
9. **Characteristic ontogenetic pace and sequencing:** Modules emerge and mature according to a species-typical developmental timetable, **largely triggered rather than learned**.

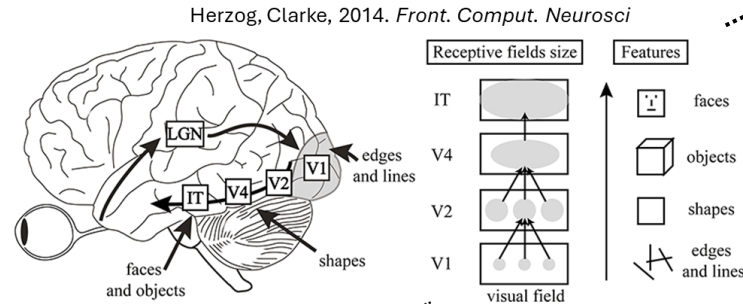
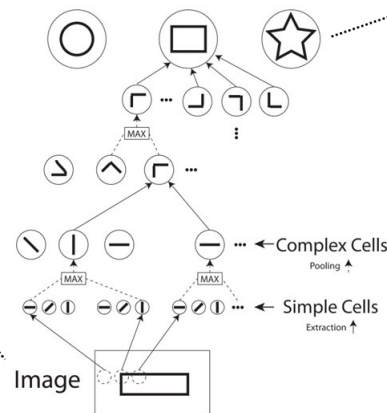
Modularity of vision

- **Input systems** (vision, audition, language) **are modular**; **central systems** (belief, reasoning) **are not**.
- **Each module performs a specialized operation in a largely feedforward sequence**
Vision: retina → V1 → V2 → inferotemporal cortex
- **Information processing relies only on proprietary databases + immediate sensory input** (encapsulation).
Early visual modules compute edges, shapes, depth, and pass representations forward.
- **Higher cognition cannot alter early visual computations.**
- **Information processing is separable**: fixed roles, primarily bottom-up flow.

Marr & bottom-up vision

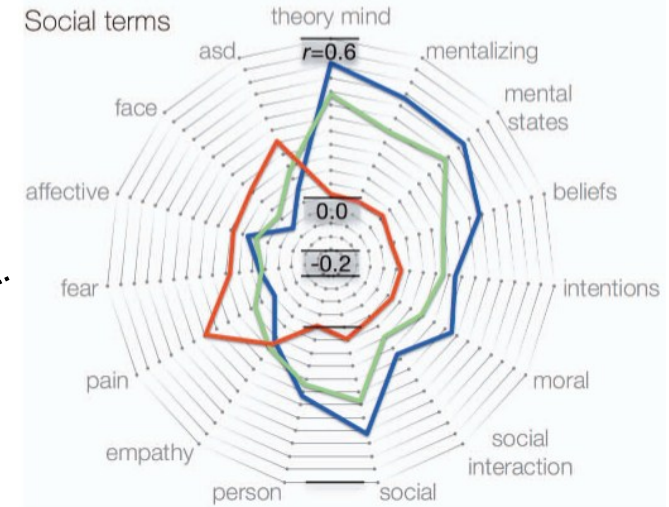
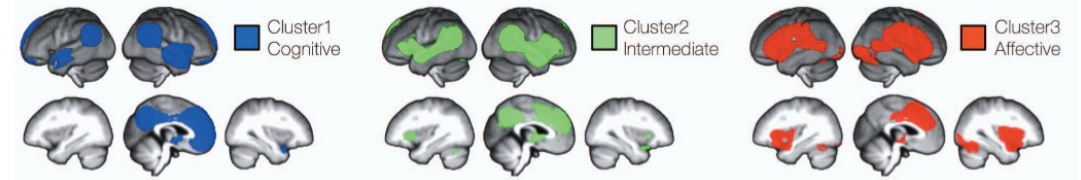
“An overall framework for visual information processing [...]:

- (1) the primal sketch, [...]
- (2) the 2 ½ D sketch, [...] and
- (3) the 3-D model representation, whose important features are that [...] primitives of various sizes are included, arranged in a modular, hierarchical organization.” (Marr, 1982)



Herzog, Clarke, 2014. *Front. Comput. Neurosci*

Schurz et al., 2021. *Psychological Bulletin*

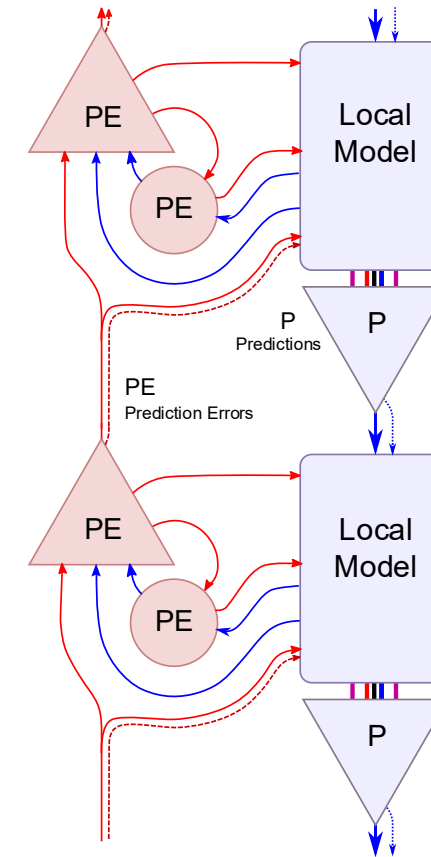


“There is some irony in this because Marr’s emphasis on vision as a bottom-up process heavily influenced opponents of the idea that perception is cognitively penetrable” (Lupyan, 2015)

Predictive coding paradigm

- Initially a model of sensory system
- Brain as a **hierarchical prediction machine**:
 - Higher levels generate expectations about what lower levels should sense;
 - Lower levels return only prediction errors (mismatch expected-actual input).
- Perception is an ongoing cycle of prediction and correction
Each level predicts the one below and updates based on its error signals

Predictive Coding
Conceptual Schematic



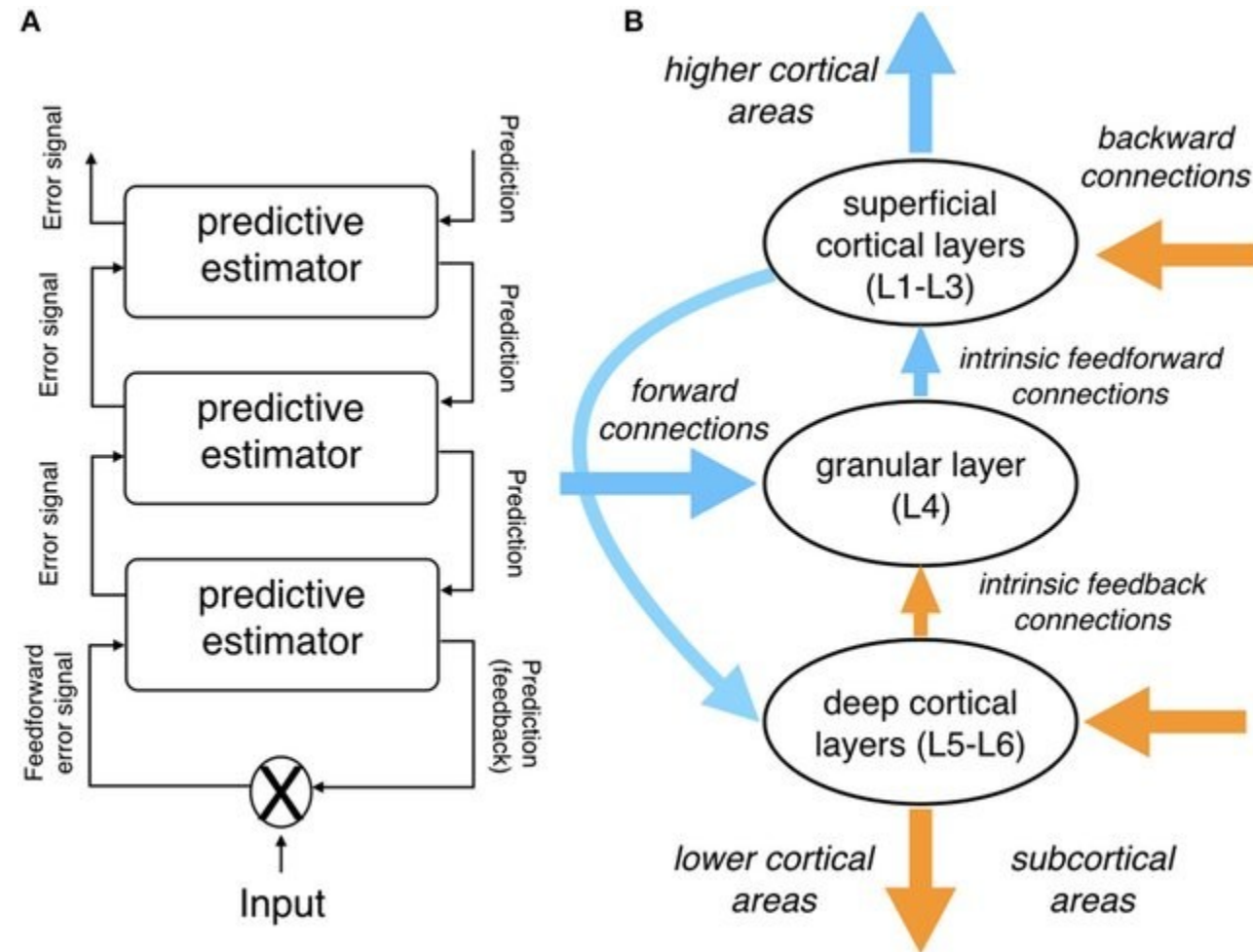
Predictive Coding compares data coming from the senses against expectations coming from the brain. When they do not match, Prediction Errors (PE) are generated.

Predictions (P) are generated by models constructed from prior experience with the world. Precision expectations flow with the predictions (dotted blue line).

Prediction Error Neurons include excitatory neurons (triangle) and inhibitory neurons (circle). A precision metric of the error is included in the signal (dotted line).

Corollaries of predictive coding hypothesis

- The **brain continuously performs hypothesis testing**: it infers causes of input and revises its model when errors appear (active inference)
- **Neural architecture is hierarchical/layered** (low → mid → high)
- **Information flows top-down and bottom-up** (continuous feedback loops)
- **Priors, expectations, task demands determine** how strongly predictions **constrain** early processing
- **Attention and uncertainty weight which signals dominate**: precise prediction errors override priors, while imprecise ones are down-weighted
- Prediction errors drive inference and learning

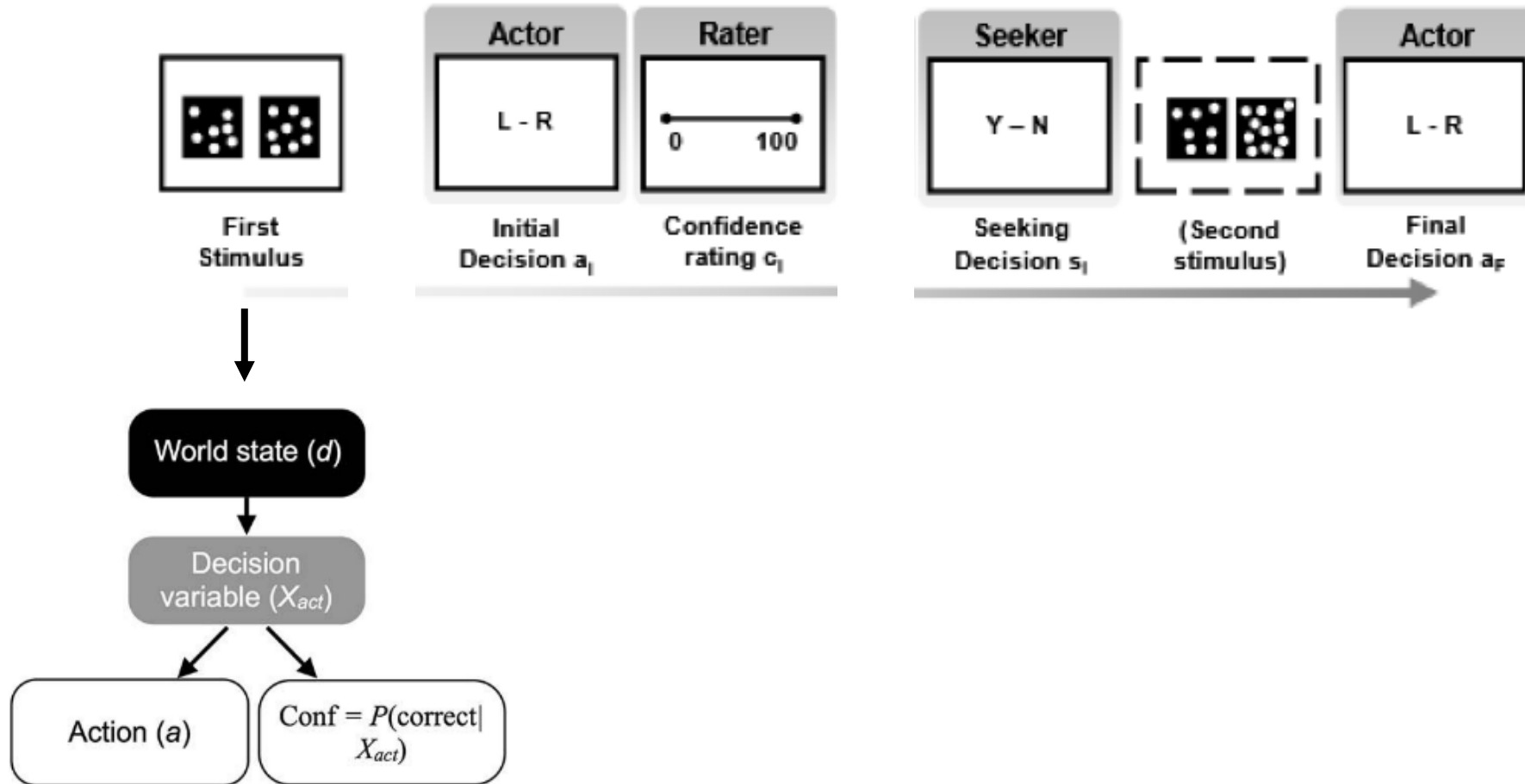


The predictive brain

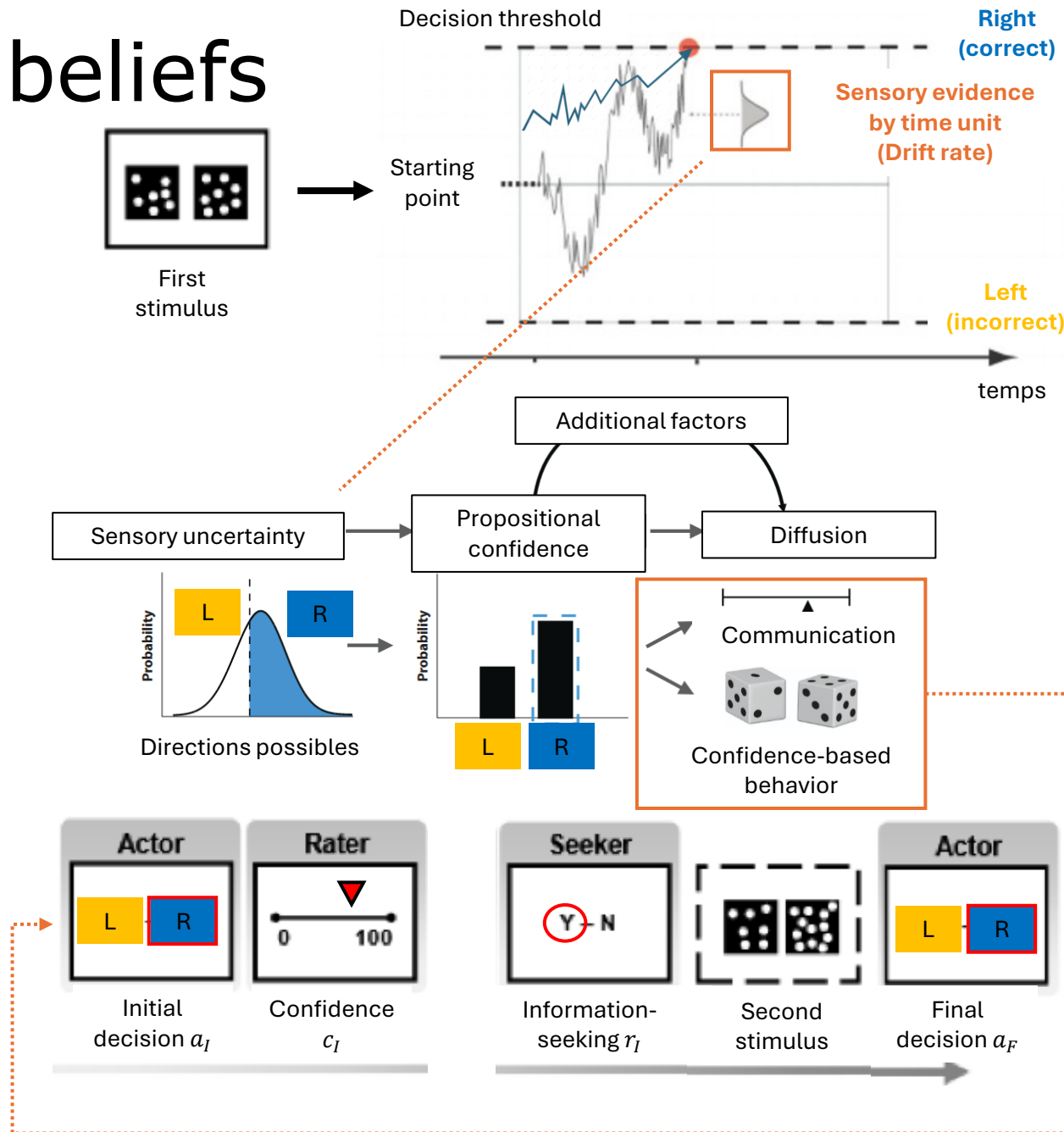
Predictive coding is member of a wider set of theories that follow the Bayesian brain hypothesis:

- Predictive coding (Sensory coding and perception)
- Bayesian inference (Brain holds probabilistic beliefs about the true state)
- Free energy principle (Minimizing surprise to maintaining adaptive states)
- Active inference (Perception-action)
- Tentatives to unify predictive coding, active inference and approximate Bayesian inference

Beliefs during decisions about percepts



Propositional beliefs



Modularity vs Predictive coding: qualitative difference

« *If a mismatch between bottom-up sensory data and top-down predictions is more effectively resolved at a higher level, then conflicts can be resolved at these higher levels leaving lower-level representations less affected. No one needs to “decide” where the influence of a higher-level state should terminate. **This is an emergent property of a hierarchical predictive system,** examples of simply performing computations that reduce global prediction error, much as water finds the shortest path downhill.* » (Lupyan, 2015)

Modularity

- Information flow fixed by architecture
- Higher-level effects occur only where explicitly wired
- Encapsulation defines boundaries of influence

Predictive Processing

- Influence emerges from prediction-error minimization
- No module “decides” where influence stops
- Global error reduction dictates how priors and errors interact